Large Data Set: Edexcel

Questions 1–17

Question 1. State the following facts about the Large Data Set:

a) The 5 locations in the UK.
b) The 3 locations not in the UK.
c) The years in which the data was collected.
d) The months which are included and the maximum total number of days in those months.

Question 2. Which of the locations

a) are coastal?
b) are south of the equator?

Question 3. Why does Perth have a lower temperature than Jacksonville for the data in the Large Data Set?

Question 4. Which of the UK locations has the highest windspeed?

Question 5. State the units of the following:

a) Daily mean temperature.
b) Daily total rainfall.
c) Daily total sunshine.
d) Daily maximum relative humidity.
e) Daily mean windspeed.
f) Cloud cover.
g) Visibility.
h) Pressure.

Question 6. Convert the following:

a) 3 knots into mph.
b) 5.75 mph into knots.
c) 30 mph into knots.

Question 7. Explain what it means for cloud coverage to be

a) 1 oktas

b) 4 oktas

c) 8 oktas

Question 8. According to the Large Data Set, on 29th May 1987, the wind in Heathrow was blowing with bearing $290°$. In which direction was the wind blowing as a compass direction?

Question 9. In the Large Data Set, is a location more likely to be high cloud coverage or low cloud coverage?

Question 10. Zara has calculated the mean and standard deviation of a sample of rainfall in Leeming in September 2015.
She used only the data which has a value, obtaining $\bar{x} = 1.6$ and $\sigma = 4.2$ where $n = 24$

a) Comment on the validity of her results.

b) Comment on how the mean and standard deviation will change after a correction.

Question 11. Two months' worth of data for hours of daily sunshine, x has been summarised for Hurn for July & August 2015 below:

$$\sum x = 321.2, \quad \sum x^2 = 2692.66, \quad n = 62$$

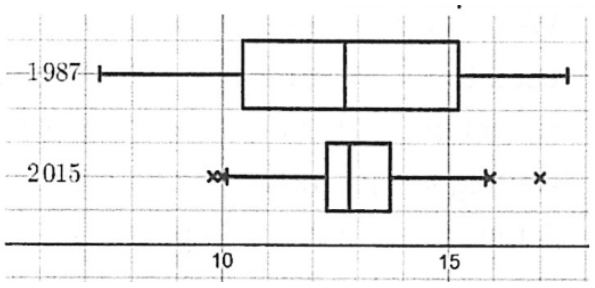
a) Calculate the mean and the standard deviation, where $s_x = \sqrt{\dfrac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}}$, giving your answer to 2 decimal places.

In Hurn for the same months in 1987 gives

$$\bar{x} = 6.42 \text{ and } s_x = 4.62$$

b) Compare these to the statistics calculated for 2015 above.

Question 12. The daily temperature for the same two consecutive months in both 1987 and 2015 in Camborne are shown in the boxplots below:



a) Which are these months likely to be?

It is claimed based on these data that the mean temperature is becoming more consistent across the UK.

b) Explain the validity of this claim.

Question 13. Jeremy is investigating the daily total rainfall in Jacksonville in October 2015. He selects data by selecting the first data point, then picks every 6th one after that until he reaches the end of the data set. State

a) State the type of sampling Jeremy used.

b) State a mistake with his application of this sampling method.

Question 14. The table below shows the probability distribution for c, the cloud coverage in Leeming in 2015.

c	1	2	3	4
$\mathrm{P}(X=c)$		0.04	0.09	0.14
c	5	6	7	8
$\mathrm{P}(X=c)$	0.17	0.22	0.23	

a) Find the missing probabilities given it is known that

$$10\mathrm{P}(c=1)=\mathrm{P}(c=8)$$

b) Find the expected number of days that the cloud coverage is 3 oktas or less.

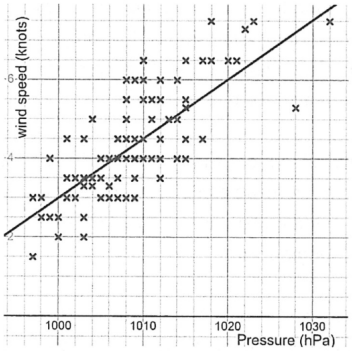
It is claimed that this data could be modelled by $X \sim \mathrm{B}\left(54, \frac{1}{9}\right)$

c) Using this model, find the number of days that cloud coverage is 3 oktas or less.

d) Comment on the validity of this claim.

Question 15. A sample of 100 data points has been taken from Beijing in 1987. The windspeed w (knots) has been plotted against pressure p (hPa) and is shown in the diagram below. A regression line has been found, given by

$$w = 0.151p - 148$$

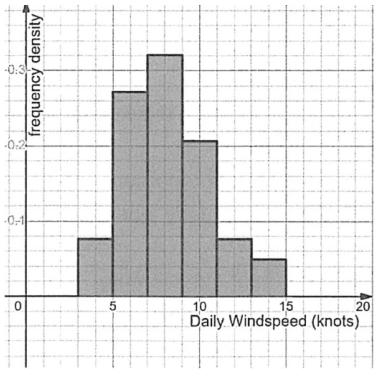


a) Use the model to predict the windspeed when there is a pressure of 1026 hPa.

The PMCC is found to be $r = 0.734$

b) Comment on why there is reason to believe that the data may not have been sampled randomly.

Question 16. Prerequisite knowledge for Q16: Normal Distribution The daily windspeed in Perth in 2015 is shown on the histogram below:



The heights of the bars are $\{0.076, 0.272, 0.321, 0.206, 0.076, 0.049\}$

a) Estimate the mean and the standard deviation of the data.

b) Estimate the number of days the windspeed is above 12 knots.

c) By modelling the data using a normal distribution with your mean and standard deviation, find the probability of the windspeed being over 12 knots.

d) Is the normal distribution a suitable model for these data?

Question 17. Prerequisite knowledge for Q17: PMCC Hypothesis Testing Adrian believes that there is a negative correlation between the windspeed and relative humidity in Leeming in 2015.
He collects a random sample of 20 dates and calculates the sample's correlation coefficient to be -0.3613
The critical value for $n = 20$ for a one-tailed test at a 5% significance level is 0.3783.

a) Carry out a hypothesis test at the 5% significance level to investigate whether Adrian's belief is supported.
b) A second sample of 20 dates is taken, and the correlation coefficient is -0.6234. Write the conclusion of the hypothesis test for this sample.